Teorema da Probabilidade Total

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup \ldots \cup (A_n \cap B)$$

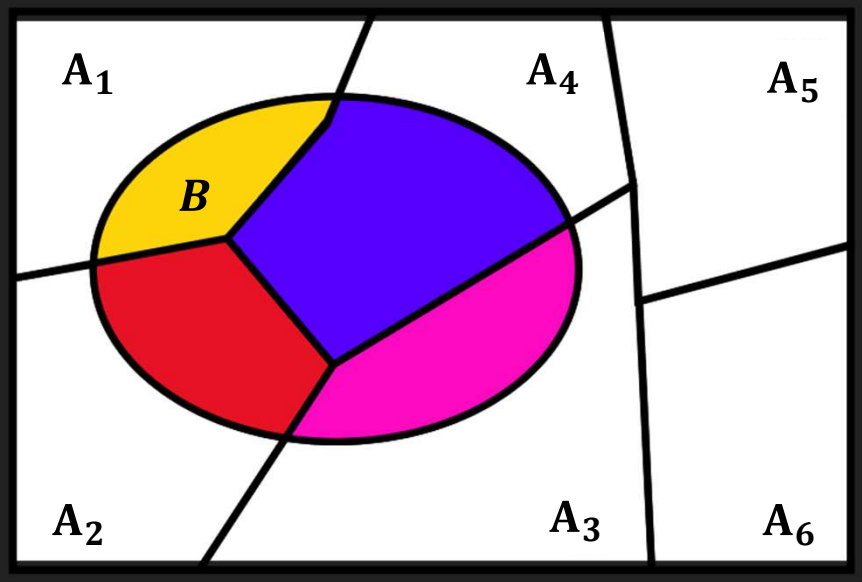
$$P(B) = P((A_1 \cap B) \cup (A_2 \cap B) \cup \ldots \cup (A_n \cap B))$$

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \ldots + P(A_n \cap B)$$

$$P(B) = \sum_{i=1}^{n} P(A_i \cap B)$$

$$P(B) = \sum_{i=1}^{n} P(B|A_i) \cdot P(A_i)$$

→ Teorema da Probabilidade Total



$$P(A_i \cap B) = P(B|A_i) \cdot P(A_i)$$

Teorema da Probabilidade Total

Seja $\{A_1, A_2, \dots, A_n\}$ uma partição do espaço amostral S, ou seja, $A_i \cap A_j = \emptyset$ para todo $1 \leq i \neq j \leq n$ e $A_1 \cup A_2 \cup \cdots \cup A_n = S$, então

$$P(B) = \sum_{i=1}^{n} P(A_i \cap B) \quad , ou\ seja , \quad P(B) = \sum_{i=1}^{n} P(B|A_i).P(A_i)$$

